Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Eddie Silveira Family Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
12081 S Cherokee RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1910

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X090-X033-XXXX

B. OPERATORS

Silveira, Eddie

Operator name: Silveira, Eddie

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
12081 S Cherokee RD	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Silveira, Eddie

Legal owner name: Silveira, Eddie

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
12081 S Cherokee RD	Gustine	CA	95322

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	650	650	250	0
Number under roof	2,200	300	0	0	0	0
Maximum number	2,200	300	650	650	250	0
Average number	2,200	300	650	650	250	0
Avg live weight (lbs)	1,400	1,450	900	650		

Predominant milk cow breed: Holstein

Average milk production: 83 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 77,689.11 *tons per reporting period*

Total nitrogen from manure: 972,616.78 *lbs per reporting period*

After ammonia losses (30% loss applied): 680,831.75 *lbs per reporting period*

Total phosphorus from manure: 161,988.96 *lbs per reporting period*

Total potassium from manure: 440,110.50 *lbs per reporting period*

Total salt from manure: 1,104,855.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 69,264,000 *gallons*

Total nitrogen generated: 99,323.66 *lbs*

Total phosphorus generated: 22,272.06 *lbs*

Total potassium generated: 133,582.35 *lbs*

Total salt generated: 984,175.92 *lbs*

	69,264,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	69,264,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #1	Ground water
Ag Well #2	Ground water
Ag Well #4	Ground water
Canal Water	Surface water
Domestic Well #1 Milk Barn	Ground water

Source Description	Type
Domestic Well #2 & #3 Tide	Ground water
Domestic Well #4 Pasture	Ground water
Domestic Well #5 Shop	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

Source Description
Tile Well

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/18/2023	Liquid commercial fertilizer *UN32*	9,080.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	24,247.23	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	24,247.23	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	Corral solids	3,240.00 *ton*	Dry-weight	21.9		33,700.00	5,310.00	20,200.00		30.10
09/15/2023	Corral solids	2,880.00 *ton*	Dry-weight	21.9		33,700.00	5,310.00	20,200.00		30.10

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	322,153.13	50,760.63	193,100.69	2,877,391.44
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	322,153.13	50,760.63	193,100.69	2,877,391.44

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 30	24	24	0	none	X070-X009-X032-XXXX
Field 31	24	24	0	none	X070-X009-X032-XXXX
Field 34	15	15	0	none	X070-X090-X016-0000
Field 35	20	20	0	none	0070-0060-0033-0000
Field 36	50	50	2	manure	X070-X090-X016-0000
Field 37	87	87	2	manure	X070-X090-X016-0000
Field 38	186	186	2	both	0070-0112-0016-0000 X070-X112-X013-XXXX X070-X112-X014-XXXX X070-X112-X015-XXXX
Field 39	240	240	2	both	X070-X112-X013-XXXX X070-X112-X014-XXXX
Field 40	15	15	1	manure	X070-X090-X042-XXXX
Field 41	52	52	2	manure	X070-X090-X042-XXXX
Field 42	38	38	2	manure	X070-X090-X042-XXXX
Totals for areas that were used for application	668	668	13		
Totals for areas that were not used for application	83	83	0		
Land application area totals	751	751	13		

B. CROPS AND HARVESTS

Field 36

Field name: Field 36

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	950.00 *ton*	Dry-weight		65.8	13,300.00	1,900.00	11,400.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	172.85	24.69	148.15	1,442.56

Field 36

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	1,300.00 *ton*	Dry-weight		64.9	11,000.00	1,210.00	4,640.00		4.72

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	200.77	22.08	84.69	861.49

Field 37

Field name: Field 37

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 87 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	1,653.00 *ton*	Dry-weight		58.6	11,700.00	1,140.00	6,760.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	184.06	17.93	106.35	2,045.16

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 87 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	2,262.00 *ton*	Dry-weight		65.5	13,000.00	1,680.00	7,680.00		4.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	233.22	30.14	137.78	891.62

Field 38

Field name: Field 38

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 186 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	3,534.00 *ton*	Dry-weight		62.9	14,400.00	1,300.00	7,480.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	203.01	18.33	105.45	1,663.56

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 186 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	4,836.00 *ton*	Dry-weight		70.9	16,900.00	2,320.00	8,900.00		6.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	255.73	35.11	134.67	935.16

Field 39

Field name: Field 39

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 240 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	4,560.00 *ton*	Dry-weight		69.8	19,600.00	2,550.00	16,400.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	224.93	29.26	188.21	1,549.26

Field 39

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 240 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	6,240.00 *ton*	Dry-weight		70.9	18,400.00	1,540.00	7,660.00		7.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	278.43	23.30	115.91	1,157.60

Field 40

Field name: Field 40

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	285.00 *ton*	Dry-weight		65.2	12,600.00	1,990.00	11,200.00		9.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	166.62	26.32	148.11	1,294.63

Field 41

Field name: Field 41

Field 41

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	988.00 *ton*	Dry-weight		57.4	14,900.00	1,710.00	12,300.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	241.20	27.68	199.11	2,007.31

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,352.00 *ton*	Dry-weight		67.6	8,800.00	1,940.00	7,560.00		5.51

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	148.26	32.69	127.37	928.32

Field 42

Field name: Field 42

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	722.00 *ton*	Dry-weight		55.0	9,880.00	2,490.00	13,800.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	168.95	42.58	235.98	1,778.40

Field 42

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	988.00 *ton*	Dry-weight		67.5	8,400.00	1,170.00	7,170.00		4.92

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	141.96	19.77	121.17	831.48

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 36 - 10/31/2022: Oats, silage-soft dough

Field name: Field 36

Crop: Oats, silage-soft dough — Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	100.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 36 - 05/28/2023: Corn, silage

Field name: Field 36

Crop: Corn, silage — Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	184.24	29.03	110.43	1,645.57	175.00 *ton*
Application event totals		184.24	29.03	110.43	1,645.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #4	Ground water	2.48	0.00	0.00	109.05	1,188,000.00 *gal*
Canal Water	Surface water	2.79	0.00	0.00	184.46	5,816,851.92 *gal*
Application event totals		5.26	0.00	0.00	293.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Application event totals		50.00	0.00	0.00	0.00	

Field 36 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.95	0.00	0.00	327.93	10,341,070.09 *gal*
Application event totals		4.95	0.00	0.00	327.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.95	0.00	0.00	327.93	10,341,070.09 *gal*
Application event totals		4.95	0.00	0.00	327.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.02	0.00	0.00	266.44	8,402,119.45 *gal*
Application event totals		4.02	0.00	0.00	266.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.02	0.00	0.00	266.44	8,402,119.45 *gal*
Application event totals		4.02	0.00	0.00	266.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #4	Ground water	1.65	0.00	0.00	72.70	792,000.00 *gal*
Canal Water	Surface water	1.86	0.00	0.00	122.97	3,877,901.28 *gal*
Application event totals		3.51	0.00	0.00	195.67	

Field 37 - 10/31/2022: Oats, silage-soft dough

Field name: Field 37

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 37 - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	174.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 37 - 05/28/2023: Corn, silage

Field name: Field 37

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	242.02	38.13	145.07	2,161.66	400.00 *ton*
Application event totals		242.02	38.13	145.07	2,161.66	

05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	5.34	0.00	0.00	353.37	19,389,506.41 *gal*
Application event totals		5.34	0.00	0.00	353.37	

06/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Application event totals		50.00	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.80	0.00	0.00	318.03	17,450,555.77 *gal*
Application event totals		4.80	0.00	0.00	318.03	

Field 37 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.27	0.00	0.00	282.69	15,511,605.13 *gal*
Application event totals		4.27	0.00	0.00	282.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.27	0.00	0.00	282.69	15,511,605.13 *gal*
Application event totals		4.27	0.00	0.00	282.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	3.20	0.00	0.00	212.02	11,633,703.85 *gal*
Application event totals		3.20	0.00	0.00	212.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	2.14	0.00	0.00	141.35	7,755,802.57 *gal*
Application event totals		2.14	0.00	0.00	141.35	

Field 38 - 11/01/2022: Oats, silage-soft dough

Field name: Field 38

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	372.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 38 - 05/30/2023: Corn, silage

Field name: Field 38

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.09	12.02	70.84	529.13	6,552,000.00 *gal*
Canal Water	Surface water	4.99	0.00	0.00	330.57	38,779,012.83 *gal*
Application event totals		59.08	12.02	70.84	859.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.49	0.00	0.00	297.51	34,901,111.55 *gal*
Application event totals		4.49	0.00	0.00	297.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.09	12.02	70.84	529.13	6,552,000.00 *gal*
Canal Water	Surface water	4.24	0.00	0.00	280.98	32,962,160.90 *gal*
Application event totals		58.33	12.02	70.84	810.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.36	10.31	60.72	453.54	5,616,000.00 *gal*
Canal Water	Surface water	4.24	0.00	0.00	280.98	32,962,160.90 *gal*
Application event totals		50.61	10.31	60.72	734.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.36	10.31	60.72	453.54	5,616,000.00 *gal*
Canal Water	Surface water	3.75	0.00	0.00	247.93	29,084,259.62 *gal*
Application event totals		50.11	10.31	60.72	701.46	

Field 38 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.89	5.52	37.79	251.97	4,680,000.00 *gal*
Canal Water	Surface water	3.25	0.00	0.00	214.87	25,206,358.34 *gal*
Application event totals		26.13	5.52	37.79	466.84	

Field 39 - 11/02/2022: Oats, silage-soft dough

Field name: Field 39

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	480.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 39 - 06/01/2023: Corn, silage

Field name: Field 39

Crop: Corn, silage Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.88	13.31	78.43	585.82	9,360,000.00 *gal*
Canal Water	Surface water	5.03	0.00	0.00	333.05	50,412,716.68 *gal*
Application event totals		64.91	13.31	78.43	918.87	

Field 39 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.64	0.00	0.00	307.43	46,534,815.39 *gal*
Application event totals		4.64	0.00	0.00	307.43	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.88	13.31	78.43	585.82	9,360,000.00 *gal*
Canal Water	Surface water	4.26	0.00	0.00	281.81	42,656,914.11 *gal*
Application event totals		64.14	13.31	78.43	867.63	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.91	10.65	62.75	468.66	7,488,000.00 *gal*
Canal Water	Surface water	3.48	0.00	0.00	230.57	34,901,111.55 *gal*
Application event totals		51.39	10.65	62.75	699.23	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.91	10.65	62.75	468.66	7,488,000.00 *gal*
Canal Water	Surface water	3.48	0.00	0.00	230.57	34,901,111.55 *gal*
Application event totals		51.39	10.65	62.75	699.23	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.83	5.99	41.01	273.38	6,552,000.00 *gal*
Canal Water	Surface water	3.10	0.00	0.00	204.95	31,023,210.26 *gal*
Application event totals		27.93	5.99	41.01	478.34	

Field 40 - 11/03/2022: Oats, silage-soft dough

Field 40 - 11/03/2022: Oats, silage-soft dough

Field name: Field 40

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	30.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 41 - 11/03/2022: Oats, silage-soft dough

Field name: Field 41

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	104.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 41 - 06/03/2023: Corn, silage

Field name: Field 41

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	101.23	15.95	60.68	904.16	100.00 *ton*
Application event totals		101.23	15.95	60.68	904.16	

Field 41 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	5.36	0.00	0.00	354.73	11,633,703.85 *gal*
Application event totals		5.36	0.00	0.00	354.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Application event totals		50.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	5.36	0.00	0.00	354.73	11,633,703.85 *gal*
Application event totals		5.36	0.00	0.00	354.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.76	0.00	0.00	315.31	10,341,070.09 *gal*
Application event totals		4.76	0.00	0.00	315.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	3.57	0.00	0.00	236.48	7,755,802.57 *gal*
Application event totals		3.57	0.00	0.00	236.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	2.68	0.00	0.00	177.36	5,816,851.92 *gal*
Application event totals		2.68	0.00	0.00	177.36	

Field 41 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.79	0.00	0.00	118.24	3,877,901.28 *gal*
Application event totals		1.79	0.00	0.00	118.24	

Field 42 - 11/04/2022: Oats, silage-soft dough

Field name: Field 42

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	59.29	17.53	75.05	783.30	76.00 *ton*
Application event totals		59.29	17.53	75.05	783.30	

Field 42 - 06/04/2023: Corn, silage

Field name: Field 42

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	117.75	18.55	70.58	1,051.68	85.00 *ton*
Application event totals		117.75	18.55	70.58	1,051.68	

Field 42 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	6.52	0.00	0.00	431.48	10,341,070.09 *gal*
Application event totals		6.52	0.00	0.00	431.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Application event totals		50.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	6.11	0.00	0.00	404.51	9,694,753.21 *gal*
Application event totals		6.11	0.00	0.00	404.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	4.89	0.00	0.00	323.61	7,755,802.57 *gal*
Application event totals		4.89	0.00	0.00	323.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	2.44	0.00	0.00	161.81	3,877,901.28 *gal*
Application event totals		2.44	0.00	0.00	161.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.63	0.00	0.00	107.87	2,585,267.52 *gal*
Application event totals		1.63	0.00	0.00	107.87	

Field 42 - 06/04/2023: Corn, silage

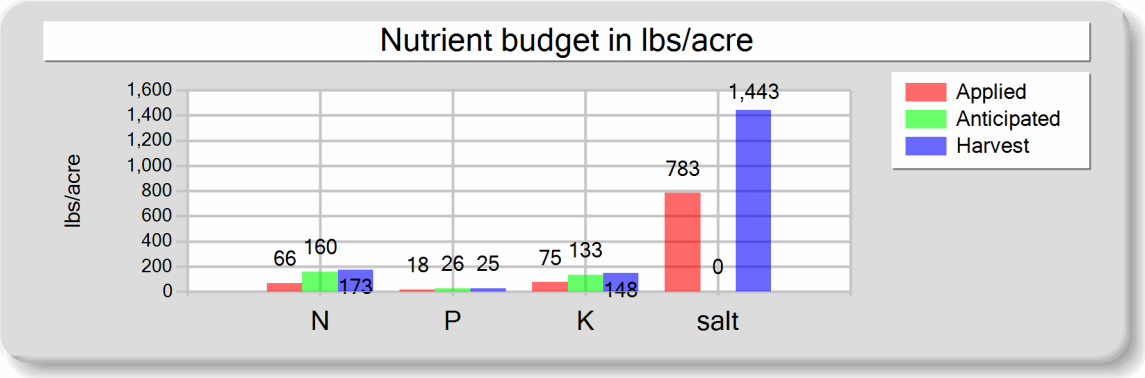
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.22	0.00	0.00	80.90	1,938,950.64 *gal*
Application event totals		1.22	0.00	0.00	80.90	

B. NUTRIENT BUDGET

Field 36 - 10/31/2022: Oats, silage-soft dough

Field name: Field 36 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	66.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.85	24.69	148.15	1,442.56
Nutrient balance	-106.55	-7.17	-73.11	-659.25
Applied to removed ratio	0.38	0.71	0.51	0.54

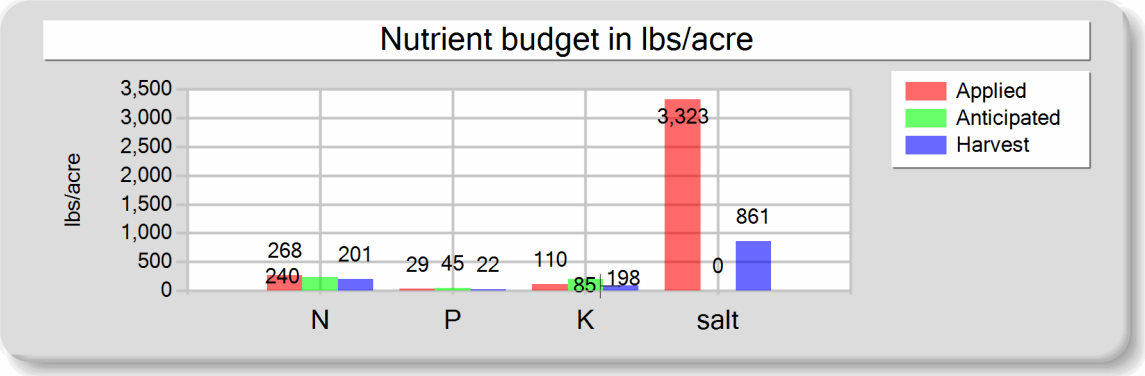
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 36 - 05/28/2023: Corn, silage

Field name: Field 36 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	184.24	29.03	110.43	1,645.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	26.73	0.00	0.00	1,677.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.97	29.03	110.43	3,323.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	200.77	22.08	84.69	861.49
Nutrient balance	67.20	6.94	25.74	2,461.99
Applied to removed ratio	1.33	1.31	1.30	3.86

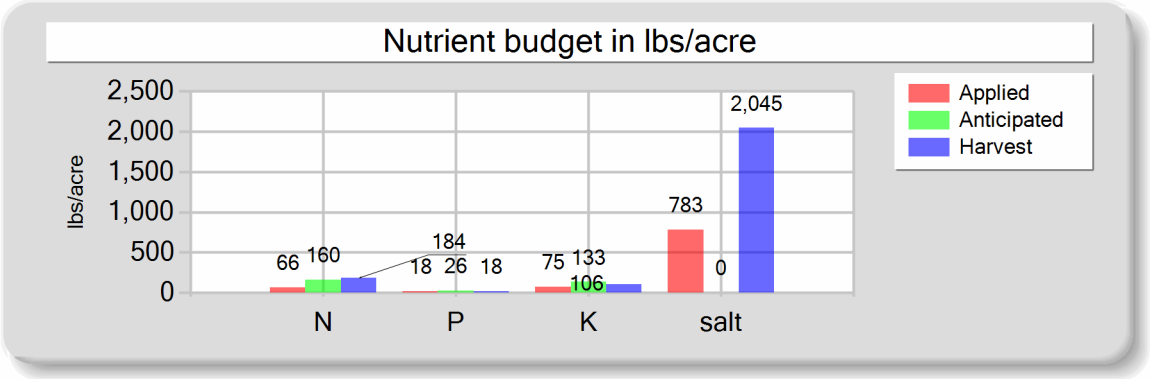
Fresh water applied	
49,161,132.28	*gallons*
1,810.44	*acre-inches*
36.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 37 - 10/31/2022: Oats, silage-soft dough

Field name: Field 37 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	66.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	184.06	17.93	106.35	2,045.16
Nutrient balance	-117.77	-0.41	-31.30	-1,261.86
Applied to removed ratio	0.36	0.98	0.71	0.38

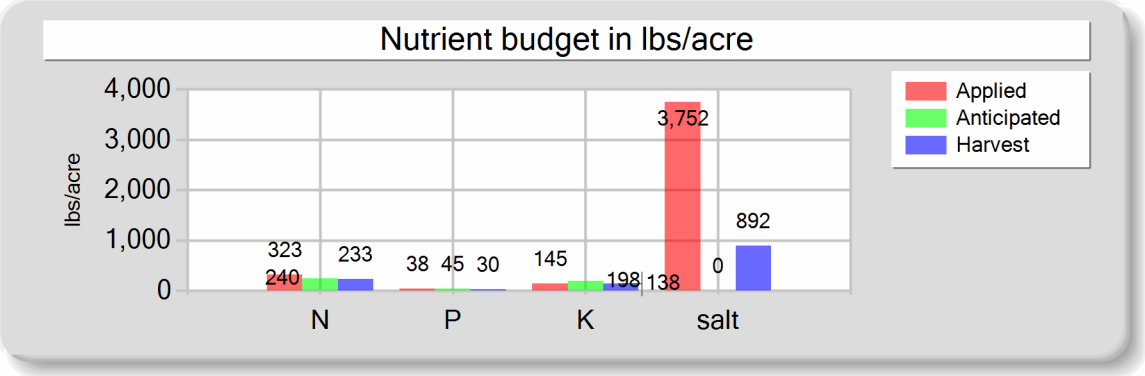
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 37 - 05/28/2023: Corn, silage

Field name: Field 37 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	242.02	38.13	145.07	2,161.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	24.02	0.00	0.00	1,590.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.04	38.13	145.07	3,751.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	233.22	30.14	137.78	891.62
Nutrient balance	89.82	8.00	7.29	2,860.20
Applied to removed ratio	1.39	1.27	1.05	4.21

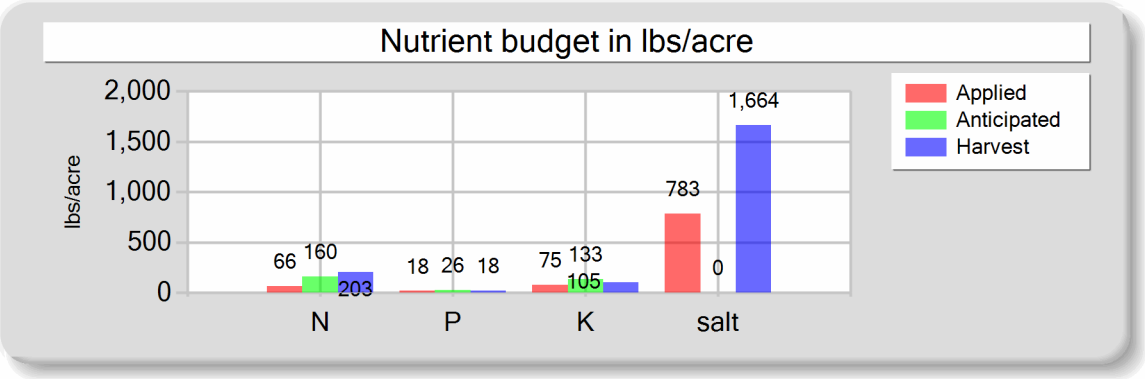
Fresh water applied	
87,252,778.86	*gallons*
3,213.22	*acre-inches*
36.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 38 - 11/01/2022: Oats, silage-soft dough

Field name: Field 38 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	66.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	203.01	18.33	105.45	1,663.56
Nutrient balance	-136.72	-0.80	-30.40	-880.26
Applied to removed ratio	0.33	0.96	0.71	0.47

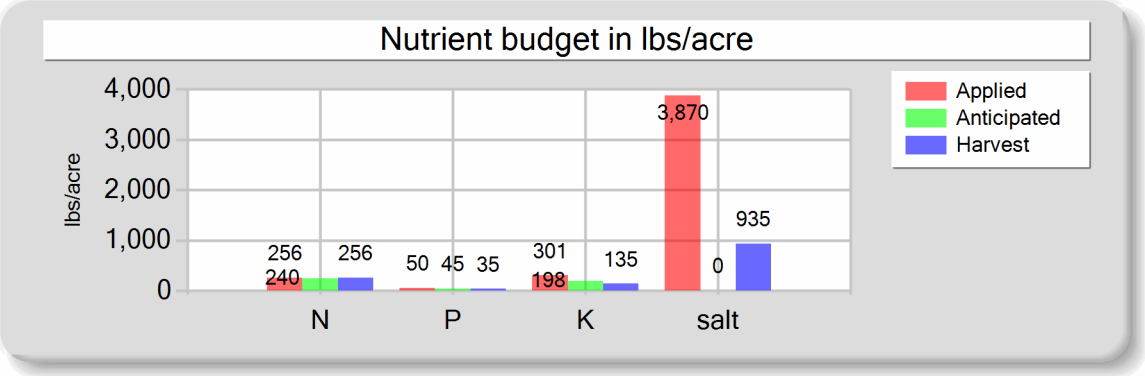
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 38 - 05/30/2023: Corn, silage

Field name: Field 38 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.79	50.18	300.93	2,217.29
Fresh water	24.97	0.00	0.00	1,652.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.75	50.18	300.93	3,870.14
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	255.73	35.11	134.67	935.16
Nutrient balance	0.02	15.07	166.26	2,934.99
Applied to removed ratio	1.00	1.43	2.23	4.14

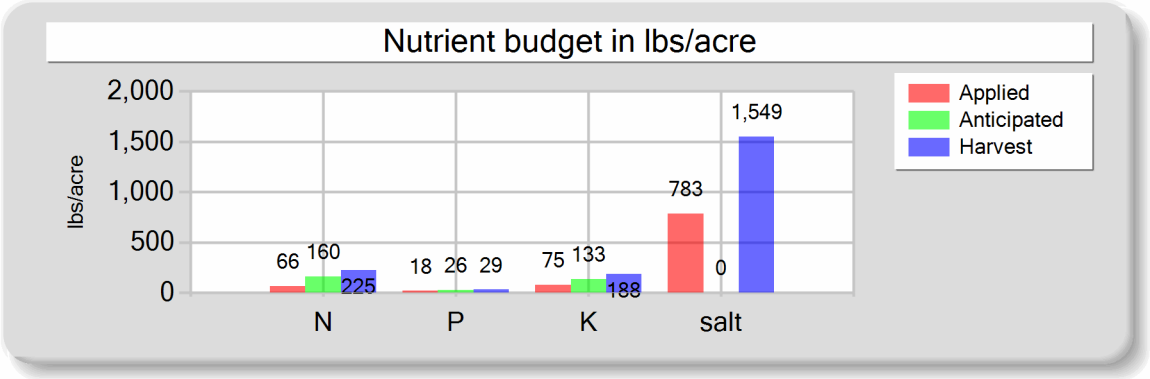
Fresh water applied	
193,895,064.14	*gallons*
7,140.50	*acre-inches*
38.39	*inches/acre*

Process wastewater applied	
29,016,000.00	*gallons*
1,068.56	*acre-inches*
5.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 39 - 11/02/2022: Oats, silage-soft dough

Field name: Field 39 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	66.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	224.93	29.26	188.21	1,549.26
Nutrient balance	-158.63	-11.74	-113.16	-765.96
Applied to removed ratio	0.29	0.60	0.40	0.51

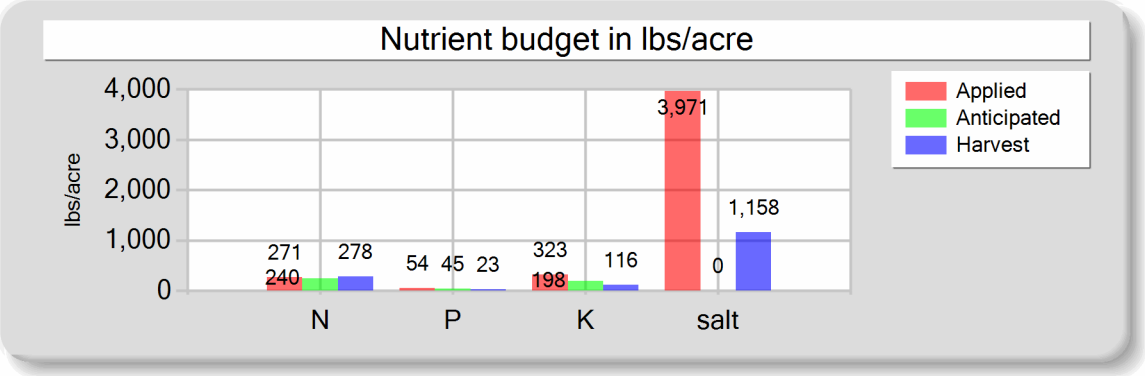
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 39 - 06/01/2023: Corn, silage

Field name: Field 39 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.41	53.91	323.37	2,382.33
Fresh water	23.99	0.00	0.00	1,588.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.41	53.91	323.37	3,970.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	278.43	23.30	115.91	1,157.60
Nutrient balance	-7.02	30.61	207.46	2,813.12
Applied to removed ratio	0.97	2.31	2.79	3.43

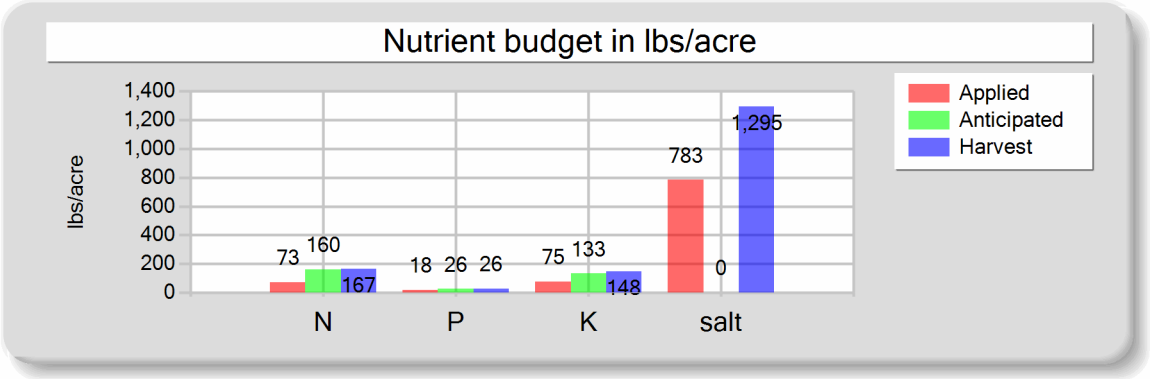
Fresh water applied	
240,429,879.54	*gallons*
8,854.21	*acre-inches*
36.89	*inches/acre*

Process wastewater applied	
40,248,000.00	*gallons*
1,482.20	*acre-inches*
6.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 40 - 11/03/2022: Oats, silage-soft dough

Field name: Field 40 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	73.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	166.62	26.32	148.11	1,294.63
Nutrient balance	-93.33	-8.79	-73.06	-511.33
Applied to removed ratio	0.44	0.67	0.51	0.61

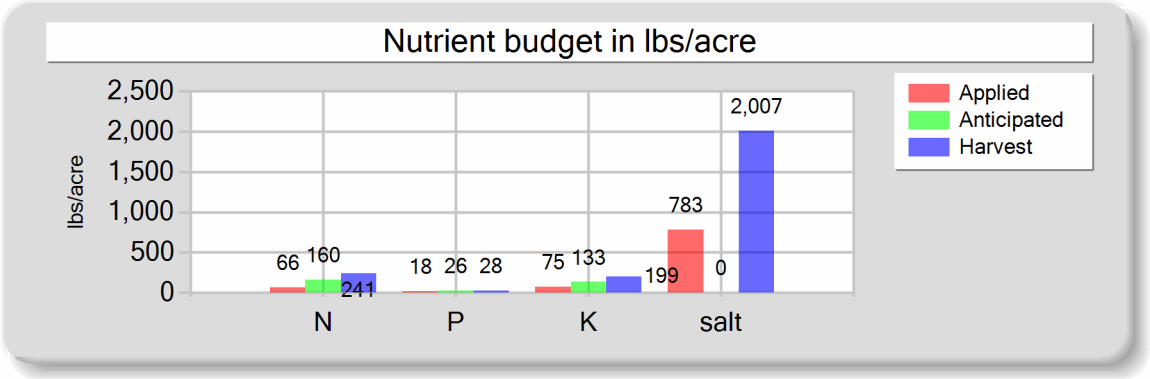
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 41 - 11/03/2022: Oats, silage-soft dough

Field name: Field 41 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	66.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	241.20	27.68	199.11	2,007.31
Nutrient balance	-174.91	-10.16	-124.06	-1,224.01
Applied to removed ratio	0.27	0.63	0.38	0.39

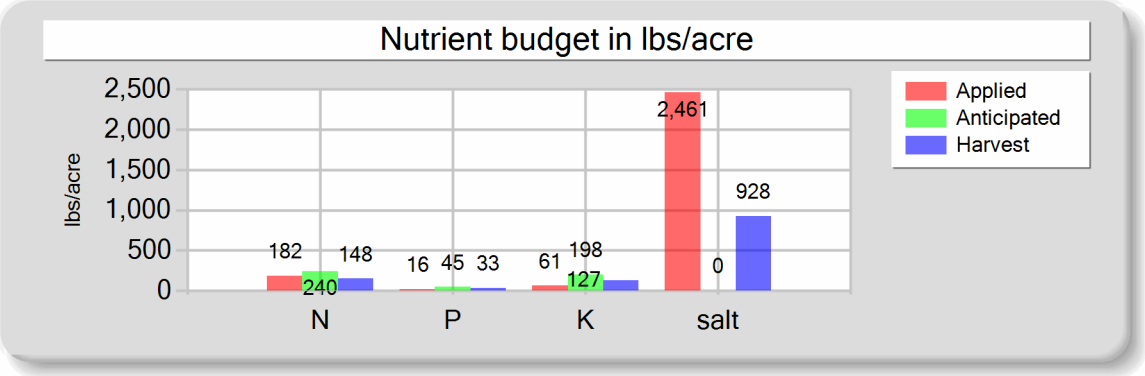
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 41 - 06/03/2023: Corn, silage

Field name: Field 41 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	101.23	15.95	60.68	904.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	23.52	0.00	0.00	1,556.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.75	15.95	60.68	2,461.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	148.26	32.69	127.37	928.32
Nutrient balance	33.48	-16.73	-66.69	1,532.69
Applied to removed ratio	1.23	0.49	0.48	2.65

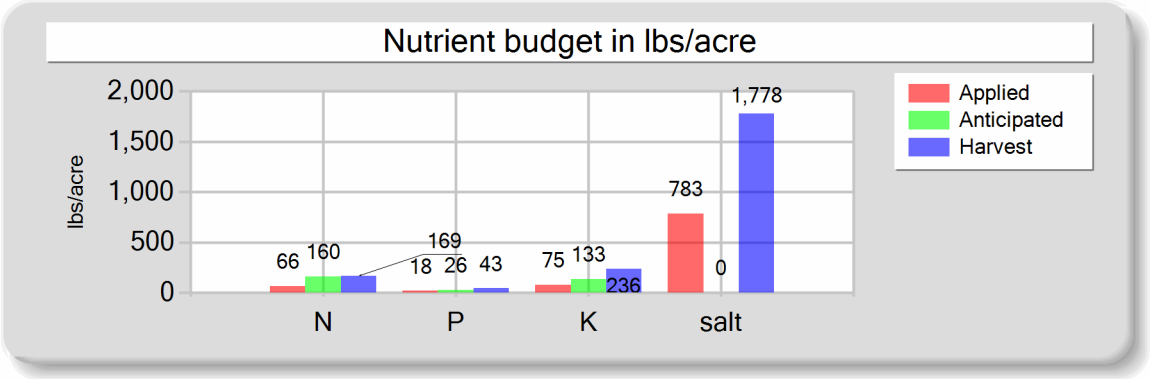
Fresh water applied	
51,059,033.56	*gallons*
1,880.33	*acre-inches*
36.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 42 - 11/04/2022: Oats, silage-soft dough

Field name: Field 42 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59.29	17.53	75.05	783.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	66.29	17.53	75.05	783.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	168.95	42.58	235.98	1,778.40
Nutrient balance	-102.65	-25.05	-160.93	-995.10
Applied to removed ratio	0.39	0.41	0.32	0.44

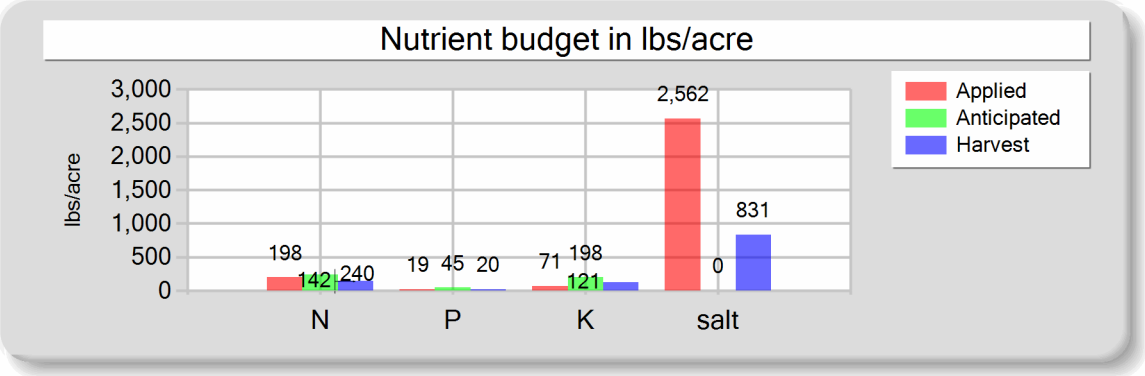
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 42 - 06/04/2023: Corn, silage

Field name: Field 42 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	50.00	0.00	0.00	0.00
Dry manure	117.75	18.55	70.58	1,051.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.81	0.00	0.00	1,510.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.56	18.55	70.58	2,561.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	141.96	19.77	121.17	831.48
Nutrient balance	55.60	-1.22	-50.60	1,730.38
Applied to removed ratio	1.39	0.94	0.58	3.08

Fresh water applied	
36,193,745.31	*gallons*
1,332.89	*acre-inches*
35.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2102010-01

Sample and source description: Manure 2102010-01

Sample date: 10/19/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,100.00	8,010.00	34,300.00	22,000.00	11,000.00	12,000.00	4,360.00	10,300.00		35.80
DL	2.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

Manure 2305490-01

Sample and source description: Manure 2305490-01

Sample date: 05/30/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 21.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,700.00	5,310.00	20,200.00							30.10
DL	5.00	5.00	5.00							0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302057-01

Sample and source description: Lagoon Water 2302057-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	505.00	505.00			66.90	513.00								4,770.00	2,800
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303395-01

Sample and source description: Lagoon Water 2303395-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	427.00	427.00			0.18	45.60								4,560.00	2,400
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2307230-01

Sample and source description: Lagoon Water 2307230-01

Sample date: 07/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	184.00	165.00			40.90	241.00								3,140.00	1,800
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310015-01

Sample and source description: Lagoon Water 2310015-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	109.00	93.00			26.30	180.00								2,050.00	1,200
DL	0.60	0.60			0.10	1.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #1

Ag Well #1

2311137-01

Sample description: 2311137-01

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.37										883.00	4,600
DL	1.40										1.00	1

Ag Well #2

2311137-02

Sample description: 2311137-02

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.50										1,020.00	560
DL	1.40										1.00	1

Ag Well #4

2311137-03

Sample description: 2311137-03

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.50										943.00	550
DL	1.40										1.00	1

Canal Water

Canal Water

2310014-01

Sample description: 2310014-01

Sample date: 09/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.87										402.00	190
DL	0.40										1.00	1

Domestic Well #1 Milk Barn

2312549-01

Sample description: 2312549-01

Sample date: 12/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			5.68								1,050.00	
DL			0.40								1.00	

Domestic Well #2 & #3 Tide

2312549-02

Sample description: 2312549-02

Sample date: 12/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			12.40								1,030.00	
DL			0.40								1.00	

Domestic Well #4 Pasture

Domestic Well #4 Pasture

2312549-03

Sample description: 2312549-03

Sample date: 12/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			7.75								933.00	
DL			0.40								1.00	

Domestic Well #5 Shop

2312549-04

Sample description: 2312549-04

Sample date: 12/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			14.20								1,140.00	
DL			0.40								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 36 - 10/31/2022: Oats, silage-soft dough

Field 36 - 10/31/2022: Oats, silage-soft dough

2305492-01

Sample and source description: 2305492-01

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	1,900.00	11,400.00		11.10
DL	0.50	0.50	0.50		0.10

Field 36 - 05/28/2023: Corn, silage

2310494-01

Sample and source description: 2310494-01

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	1,210.00	4,640.00		4.72
DL	0.50	0.50	0.50		0.10

Field 37 - 10/31/2022: Oats, silage-soft dough

2305492-02

Sample and source description: 2305492-02

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	1,150.00	6,700.00		11.00
DL	0.50	0.50	0.50		0.10

Field 37 - 10/31/2022: Oats, silage-soft dough

2305492-03

Sample and source description: 2305492-03

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	1,140.00	6,760.00		13.00
DL	0.50	0.50	0.50		0.10

Field 37 - 05/28/2023: Corn, silage

2310494-02

Sample and source description: 2310494-02

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	1,680.00	7,680.00		4.97
DL	0.50	0.50	0.50		0.10

2310494-03

Sample and source description: 2310494-03

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	1,430.00	7,090.00		5.00
DL	0.50	0.50	0.50		0.10

Field 38 - 11/01/2022: Oats, silage-soft dough

Field 38 - 11/01/2022: Oats, silage-soft dough

2305492-04

Sample and source description: 2305492-04

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,180.00	1,840.00	12,500.00		10.10
DL	0.50	0.50	0.50		0.10

2305492-05

Sample and source description: 2305492-05

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	1,300.00	7,480.00		11.80
DL	0.50	0.50	0.50		0.10

2305492-06

Sample and source description: 2305492-06

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,920.00	1,280.00	8,350.00		10.10
DL	0.50	0.50	0.50		0.10

Field 38 - 11/01/2022: Oats, silage-soft dough

2305492-07

Sample and source description: 2305492-07

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,740.00	1,510.00	7,720.00		12.90
DL	0.50	0.50	0.50		0.10

Field 38 - 05/30/2023: Corn, silage

2310494-04

Sample and source description: 2310494-04

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	1,770.00	8,600.00		6.07
DL	0.50	0.50	0.50		0.10

2310494-05

Sample and source description: 2310494-05

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	2,490.00	11,300.00		6.42
DL	0.50	0.50	0.50		0.10

Field 38 - 05/30/2023: Corn, silage

2310494-06

Sample and source description: 2310494-06

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	2,320.00	8,900.00		6.18
DL	0.50	0.50	0.50		0.10

2310494-07

Sample and source description: 2310494-07

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	2,520.00	9,170.00		6.74
DL	0.50	0.50	0.50		0.10

Field 39 - 11/02/2022: Oats, silage-soft dough

2305491-01

Sample and source description: 2305491-01

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	1,040.00	5,360.00		10.20
DL	0.50	0.50	0.50		0.10

Field 39 - 11/02/2022: Oats, silage-soft dough

2305491-02

Sample and source description: 2305491-02

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,360.00	1,410.00	6,700.00		11.00
DL	0.50	0.50	0.50		0.10

2305491-03

Sample and source description: 2305491-03

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,860.00	1,080.00	6,980.00		8.50
DL	0.50	0.50	0.50		0.10

2305491-04

Sample and source description: 2305491-04

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,590.00	1,860.00	13,100.00		17.20
DL	0.50	0.50	0.50		0.10

Field 39 - 11/02/2022: Oats, silage-soft dough

2305491-05

Sample and source description: 2305491-05

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	2,550.00	16,400.00		13.50
DL	0.50	0.50	0.50		0.10

2305491-06

Sample and source description: 2305491-06

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,070.00	1,440.00	9,200.00		9.92
DL	0.50	0.50	0.50		0.10

Field 39 - 06/01/2023: Corn, silage

2310493-01

Sample and source description: 2310493-01

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	16,000.00	7,030.00		6.95
DL	0.50	0.50	0.50		0.10

Field 39 - 06/01/2023: Corn, silage

2310493-02

Sample and source description: 2310493-02

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	1,540.00	7,660.00		7.65
DL	0.50	0.50	0.50		0.10

2310493-03

Sample and source description: 2310493-03

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	2,130.00	8,100.00		7.97
DL	0.50	0.50	0.50		0.10

2310493-04

Sample and source description: 2310493-04

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	3,060.00	8,140.00		7.25
DL	0.50	0.50	0.50		0.10

Field 39 - 06/01/2023: Corn, silage

2310493-04

Sample and source description: 2310493-04

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,580.00	12,900.00		6.98
DL	0.50	0.50	0.50		0.10

Field 40 - 11/03/2022: Oats, silage-soft dough

2305491-07

Sample and source description: 2305491-07

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	1,990.00	11,200.00		9.79
DL	0.50	0.50	0.50		0.10

Field 41 - 11/03/2022: Oats, silage-soft dough

2305491-08

Sample and source description: 2305491-08

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	1,710.00	12,300.00		12.40
DL	0.50	0.50	0.50		0.10

Field 41 - 06/03/2023: Corn, silage

2310493-06

Sample and source description: 2310493-06

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,800.00	1,940.00	7,560.00		5.51
DL	0.50	0.50	0.50		0.10

Field 42 - 11/04/2022: Oats, silage-soft dough

2305491-09

Sample and source description: 2305491-09

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,880.00	2,490.00	13,800.00		10.40
DL	0.50	0.50	0.50		0.10

Field 42 - 06/04/2023: Corn, silage

2310493-07

Sample and source description: 2310493-07

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,980.00	1,710.00	10,200.00		5.28
DL	0.50	0.50	0.50		0.10

Field 42 - 06/04/2023: Corn, silage

2310493-08

Sample and source description: 2310493-08

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,400.00	1,170.00	7,170.00		4.92
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

Tile Well

2311137-04

Sample description: 2311137-04

Sample date: 11/12/2023 Source of analysis: Lab analysis

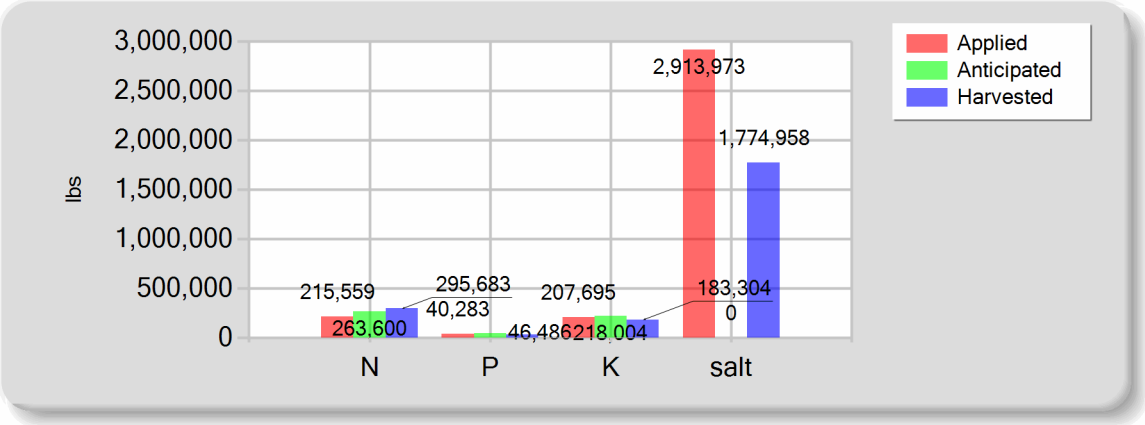
	NH4-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		36.50		1,840.00	970
DL		0.40		1.00	1

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

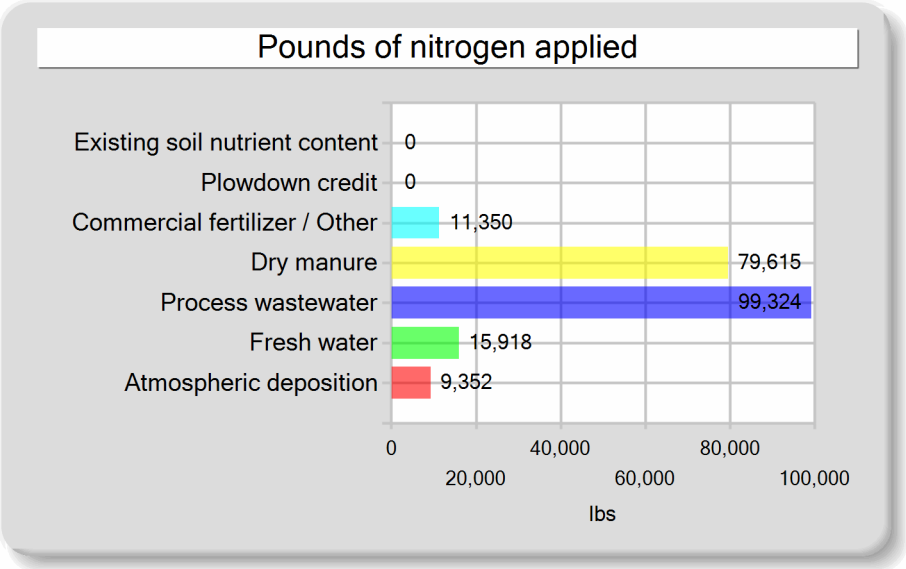
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

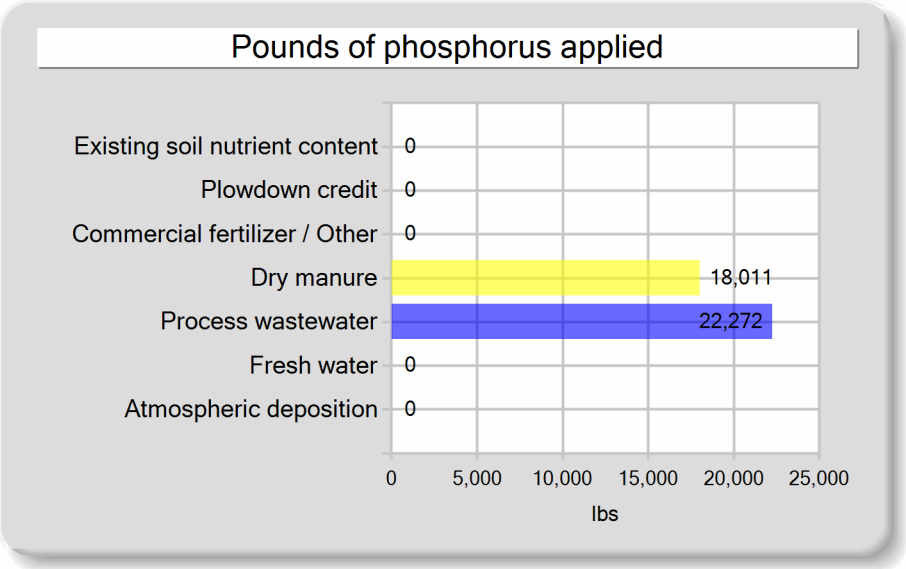
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	11,350.00	0.00	0.00	0.00
Dry manure	79,614.87	18,010.90	74,112.16	880,570.19
Process wastewater	99,323.66	22,272.06	133,582.35	984,175.92
Fresh water	15,918.12	0.00	0.00	1,049,226.95
Atmospheric deposition	9,352.00	0.00	0.00	0.00
Total nutrients applied	215,558.64	40,282.95	207,694.51	2,913,973.06
Anticipated crop nutrient removal	263,600.00	46,485.80	218,004.40	0.00
Actual crop nutrient removal	295,682.66	34,979.21	183,303.84	1,774,958.34
Nutrient balance	-80,124.01	5,303.74	24,390.67	1,139,014.72
Applied to removed ratio	0.73	1.15	1.13	1.64

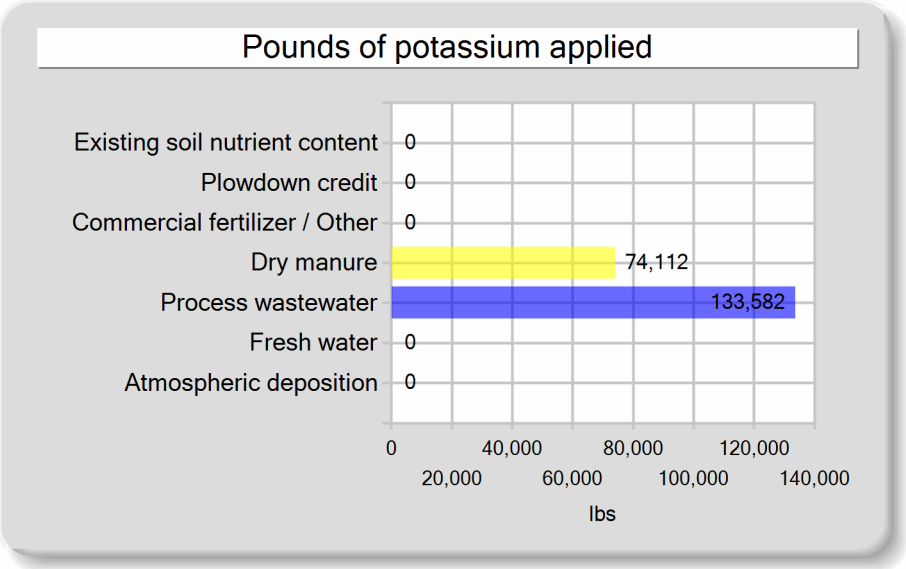
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

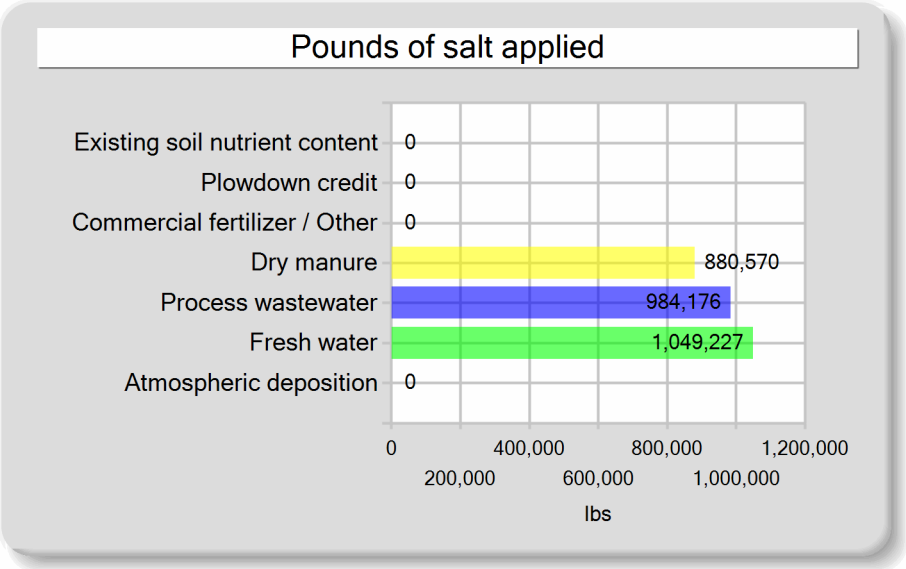


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X Eddie Silveira

SIGNATURE OF OWNER OF FACILITY

Eddie Silveira

PRINT OR TYPE NAME

✓ 7-16-24

DATE

SIGNATURE OF OPERATOR OF FACILITY

SAME AS OWNER

PRINT OR TYPE NAME

DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	See Attachments
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	